SEMINAR REPORT

A seminar Report on

ML-Based ATS Resume Analyzer and Optimizer

Submitted in partial fulfillment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

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ABSTRACT

Recruitment involves analyzing resumes and comparing candidate qualifications with job requirements. Resumes may contain information in different formats, making consistent information extraction and comparison challenging. This seminar report studies research related to resume screening, skill extraction, resume–job matching, machine learning, and large language models.

The proposed **ML-Based ATS Resume Analyzer and Optimizer** follows a four-stage workflow: Optical Character Recognition (OCR), Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs). OCR is used when text must be extracted from scanned or image-based resumes. NLP processes the extracted text and identifies relevant information, including skills, qualifications, work experience, and certifications. The ML stage compares the extracted resume information with job requirements to estimate compatibility and identify potential skill gaps. The LLM stage uses these results and supporting information to generate explanations and personalized resume-improvement suggestions.

The literature review examines ten selected research papers: one related to OCR and post-OCR processing, three related to NLP, three related to machine learning, and three related to LLM-based resume intelligence. The studies are examined in terms of their objectives, methodologies, findings, limitations, and relevance to the proposed system.

The report also presents the proposed architecture, processing workflow, evaluation approach, advantages, limitations, and project-initiation plan. Since the proposed implementation has not been evaluated within this report, its effectiveness cannot yet be established. Further testing is required to assess text-extraction quality, information-extraction performance, matching relevance, and the factual accuracy of generated feedback.

Keywords: ATS Resume Analyzer, Resume Optimization, OCR, NLP, Machine Learning, Large Language Models, Resume–Job Matching, Skill-Gap Analysis

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Chapter 1: Introduction

1.1 Introduction

Recruitment is an important process in which organizations examine resumes to identify candidates whose qualifications, skills, and experience match the requirements of a job. An Applicant Tracking System (ATS) helps organizations manage job applications and organize candidate information. However, analyzing resumes manually can be time-consuming, especially when a large number of applications are received.

Resumes contain different types of information, including personal details, educational qualifications, technical skills, work experience, projects, and certifications. This information may be presented in different formats and structures, making it difficult to compare resumes consistently with job descriptions.

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system designed to analyze a resume against a specific job description. It aims to extract relevant information, compare candidate skills with job requirements, identify potential skill gaps, and provide suggestions for improving the resume.

The proposed system integrates four technologies: Optical Character Recognition (OCR), Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs). OCR extracts text from scanned or image-based resumes when direct text extraction is unsuitable. NLP processes the extracted text and identifies important information. Machine learning supports resume–job matching, while LLMs generate explanations and personalized resume-improvement suggestions based on the available evidence.

The system aims to help job seekers understand how their resumes relate to particular job requirements and identify areas that may need improvement. However, the matching results are intended to support decision-making and do not guarantee ATS acceptance, an interview, or employment.

1.2 Background:

With the increasing use of digital recruitment platforms, organizations receive resumes in various formats, such as PDF documents, Word files, and scanned images. Reviewing these documents requires the identification of relevant qualifications, skills, experience, and other job-related information.

Traditional resume screening may depend heavily on manual review or keyword matching. Although these approaches can help identify relevant information, they may not fully capture differences in terminology or the relationship between a candidate's documented skills and the requirements of a job.

Natural Language Processing helps computers process and analyze human language. In resume analysis, NLP can be used to identify skills, qualifications, experience, and other important information from resumes and job descriptions.

Machine Learning enables systems to learn patterns from data and use them to make predictions or support decisions. In the proposed system, suitable ML methods may be used to estimate compatibility between resume information and job requirements.

Large Language Models have expanded the possibilities for generating natural-language explanations and suggestions. They can help explain matching results and suggest ways to improve the presentation of existing qualifications and experience.
However, their outputs must be checked to ensure that they do not introduce inaccurate or unsupported information.

By integrating OCR, NLP, ML, and LLMs, the proposed system aims to provide a structured approach to resume analysis and optimization.

1.3 Problem Statement

Job seekers often find it difficult to determine whether their resumes adequately address the requirements of a particular job. Resumes may contain relevant skills and qualifications, but differences in formatting, terminology, and presentation can make this information difficult to identify and compare.

Manual resume analysis can require considerable time and effort. Simple keyword-based approaches may also fail to recognize related expressions or adequately explain the differences between a resume and a job description. Furthermore, scanned resumes may require text extraction before their contents can be analysed.

Therefore, there is a need for a system that can extract relevant information from resumes, analyse it using NLP, compare it with job requirements using suitable ML methods, identify potential skill gaps, and generate understandable resume-improvement suggestions using an LLM.

The proposed system aims to address these challenges while maintaining the accuracy of the candidate's original information and allowing users to review the generated suggestions.

1.4 Aim

The aim of the project is to study and propose an **ML-Based ATS Resume Analyzer and Optimizer** that integrates OCR, NLP, Machine Learning, and Large Language Models to analyse resumes, compare them with job descriptions, identify potential skill gaps, and generate personalized resume-improvement feedback.

1.5 Objectives of the Project

The main objectives of the proposed project are:

1. To study existing research on resume parsing, skill extraction, resume screening, and resume optimization.
2. To examine OCR-based text extraction for scanned and image-based resumes.
3. To process resume text and job descriptions using NLP techniques.
4. To identify relevant skills, qualifications, experience, and other job-related information.
5. To study suitable machine learning approaches for resume–job matching.
6. To identify potential skill gaps between candidate information and job requirements.
7. To examine how LLMs can generate personalized resume-improvement suggestions.
8. To define an evaluation approach for measuring extraction quality, matching relevance, and feedback accuracy.
9. To identify the limitations, ethical considerations, and resource requirements of the proposed system.

1.6 Scope of the Project

The proposed project focuses on analysing a resume in relation to a selected job description The scope includes:

- **Resume Input:** Accepting a resume in a supported digital document or image format.
- **Text Extraction:** Extracting text directly from suitable digital documents and applying OCR when necessary.
- **NLP-Based Processing:** Processing resume text and job descriptions to identify relevant information.
- **Information Extraction:** Identifying skills, qualifications, experience, projects, and certifications.
- **ML-Based Matching:** Comparing extracted resume information with the requirements of a selected job.
- **Skill-Gap Analysis:** Highlighting requirements that appear to be missing or insufficiently documented.
- **LLM-Based Feedback:** Generating explanations and suggestions for improving the presentation of existing qualifications and experience.
- **Result Presentation:** Presenting matching information and suggestions for user review.

The project does not guarantee that a resume will pass every ATS, that a candidate will receive

an interview, or that employment will be obtained. The actual capabilities and performance of the system will depend on its implementation, available data, and evaluation results.

1.7 Technologies Used

The proposed system uses four main technology categories.

1.7.1 Optical Character Recognition (OCR)

OCR converts text in scanned documents and images into machine-readable text. It is used when the resume's text cannot be extracted directly or when direct extraction is unsuitable.

1.7.2 Natural Language Processing (NLP)

NLP processes resume text and job descriptions to identify relevant information, such as technical skills, qualifications, experience, and job requirements.

1.7.3 Machine Learning (ML)

Machine Learning supports the analysis of extracted information and the estimation of compatibility between a resume and a job description. The particular model and matching method must be selected and evaluated during implementation.

1.7.4 Large Language Models (LLMs)

LLMs generate natural-language explanations and resume-improvement suggestions based on matching results and supporting evidence. Their outputs must be checked against the original documents to avoid unsupported claims.

1.8 Organization of the Report

This report is organized into seven chapters.:

- **Chapter 1 – Introduction:** Presents the project background, problem statement, aim, objectives, scope, and technologies used.
- **Chapter 2 – Literature Review:** Reviews the selected research papers related to OCR, NLP, Machine Learning, and LLMs.
- **Chapter 3 – Information on Seminar Topic:** Explains the proposed system, its working principle, architecture, technical methods, and evaluation approach.
- **Chapter 4 – Advantages and Disadvantages:** Discusses the potential benefits and limitations of the proposed system.
- **Chapter 5 – Conclusion:** Summarizes the seminar study and identifies possible future work.
- **Chapter 6 – Project Initiation:** Describes the problem statement, scope, objectives, stakeholders, and required resources.
- **Chapter 7 – Seminar Presentation:** Presents the seminar structure and possible questions and answers.

Chapter 2: Literature Review

Review of Research Papers

1. **Resume Information Extraction via Post-OCR Text Processing**

   This research focuses on extracting information from resumes using Optical Character Recognition (OCR) and post-OCR text processing. OCR converts text from scanned documents and images into machine-readable text. Post-OCR processing helps prepare the extracted text for further analysis. This approach is useful for processing scanned resumes before identifying skills, qualifications, and experience.

2. **Technical Skills Information Extraction from Resumes Using Advanced Natural Language Processing Model with Transfer Learning**

   This research studies the extraction of technical skills from resumes using Natural Language Processing (NLP) and transfer learning. It explores methods for identifying technical skills from employment-related text. Such techniques can help organize resume information and identify skills relevant to specific job requirements.

3. **Resume Screening with Natural Language Processing (NLP)**

   This research focuses on applying NLP techniques to resume screening. NLP helps process resume text and identify relevant information, such as skills, qualifications, and work experience. These techniques can support the analysis of candidate profiles and help compare resume information with job-related requirements.

4. **Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening**

   This research examines a resume-screening approach that combines NLP with ensemble-based machine learning techniques. The approach focuses on processing resume information and using automated methods to support candidate screening. It is relevant to systems that combine text processing with machine-learning-based analysis.

5. **Resume Screening Using Machine Learning**

   This research studies the use of Machine Learning (ML) techniques for automated resume screening. Machine-learning methods can analyse information from resumes and support screening-related tasks. The selection of a suitable model depends on the available data, the screening objective, and the evaluation method.

6. **Automated Resume Screening Using Machine Learning: An Ensemble-Based Approach for Efficient Candidate Selection**

   This research examines the application of ensemble learning to automated resume screening. Ensemble learning combines multiple models or predictions to address a The

study is relevant to understanding how machine-learning approaches can support candidate screening and automated resume analysis.

7. Design and Development of Machine Learning Based Resume Ranking System

This research focuses on developing a machine-learning-based resume ranking system. Resume ranking involves ordering candidate resumes according to their relevance to a particular job or criterion. Such an approach is useful when multiple resumes need to be compared against specified job requirements.

8. Optimizing Resume Design for ATS Compatibility: A Large Language Model Approach

This research examines the use of Large Language Models (LLMs) for improving resume design and ATS compatibility. LLMs can generate natural-language suggestions that help users improve the presentation of their existing qualifications and experience. This approach is relevant to resume optimization and personalized feedback generation.

9. Resume Analyzer Using LLM

This research focuses on using Large Language Models for resume analysis. LLM-based approaches can be studied for their ability to interpret resume information and present understandable results. The specific methods and capabilities depend on the system described in the research paper.

10. An LLM-Based Resume Intelligence System for Recruiters and Job Seekers

This research examines the application of Large Language Models in resume intelligence for recruiters and job seekers. Such systems may support resume-related analysis and the presentation of job-related information. The paper is relevant to studying how language models can assist users in understanding and improving their resumes.

Sr. No.	Research Paper	Approach Type	Technology Used	Key Objective
1	Resume Information Extraction via Post-OCR Text Processing	Text extraction and processing	OCR	Extract text from scanned resumes
2	Technical Skills Information Extraction from Resumes Using Advanced NLP Model with Transfer Learning	Skill extraction	NLP and transfer learning	Identify technical skills from resumes

3	Resume Screening with Natural Language Processing (NLP)	Text-based screening	NLP	Process resume information
4	Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening	Hybrid screening	NLP and ensemble learning	Support automated resume screening
5	Resume Screening Using Machine Learning	Automated screening	Machine Learning	Study machine-learning-based screening
6	Automated Resume Screening Using Machine Learning: An Ensemble-Based Approach for Efficient Candidate Selection	Ensemble-based screening	Machine Learning	Support candidate selection
7	Design and Development of Machine Learning Based Resume Ranking System	Resume ranking	Machine Learning	Rank resumes according to job relevance
8	Optimizing Resume Design for ATS Compatibility: A Large Language Model Approach	Resume optimization	LLM	Generate resume-improvement suggestions
9	Resume Analyzer Using LLM	Resume analysis	LLM	Analyse resume information
10	An LLM-Based Resume Intelligence System for Recruiters and Job Seekers	Resume intelligence	LLM	Support resume analysis for recruiters and job seekers

Table 2.1: Literature Review

Note: Verify the research-paper titles, authors, publication details, methods, and findings against the original PDFs before submitting the report.

Chapter 3: Information On Seminar

3.1 Overview of ML-Based ATS Resume Analyzer and Optimizer

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system designed to analyse resumes and compare them with specific job descriptions. It integrates four technologies: Optical Character Recognition (OCR), Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs).

OCR extracts text from scanned or image-based resumes, while digital documents can be processed through direct text extraction. NLP techniques identify important information such as skills, educational qualifications, and work experience. Machine Learning helps compare resume information with job requirements and estimate compatibility. Finally, LLMs generate personalized suggestions to improve resume content, clarity, and relevance.

The main objective of the proposed system is to help job seekers understand how well their resumes match specific job requirements, identify potential skill gaps, and make informed improvements to their resumes.

3.2 Working Principle

The proposed system follows the steps given below.

Step 1: Resume Input

The user uploads a resume in a supported document format.

Step 2: Job Description Input

The user provides the job description containing the required skills, qualifications, and experience.

Step 3: Text Extraction

Text is extracted directly from digital documents. OCR is applied when the resume contains scanned or image-based text.

Step 4: Text Preprocessing

The extracted text is cleaned and organised for further processing. Unnecessary formatting and irrelevant elements are handled appropriately.

Step 5: NLP-Based Information Extraction

NLP techniques are used to identify important information, including technical skills, educational qualifications, and work experience.

Step 6: Job Requirement Analysis

The job description is analyzed to identify the skills, qualifications, and experience relevant to the position.

Step 7: Machine Learning-Based Matching

The extracted resume information is compared with the job requirements using a selected matching approach.

Step 8: Skill Gap Analysis

The system identifies relevant job requirements that are missing or insufficiently represented in the resume.

Step 9: LLM-Based Feedback

A Large Language Model generates suggestions for improving resume clarity, wording, and relevance to the job description.

Step 10: Result Display:

The system presents the matching results, potential skill gaps, and resume improvement suggestions to the user.

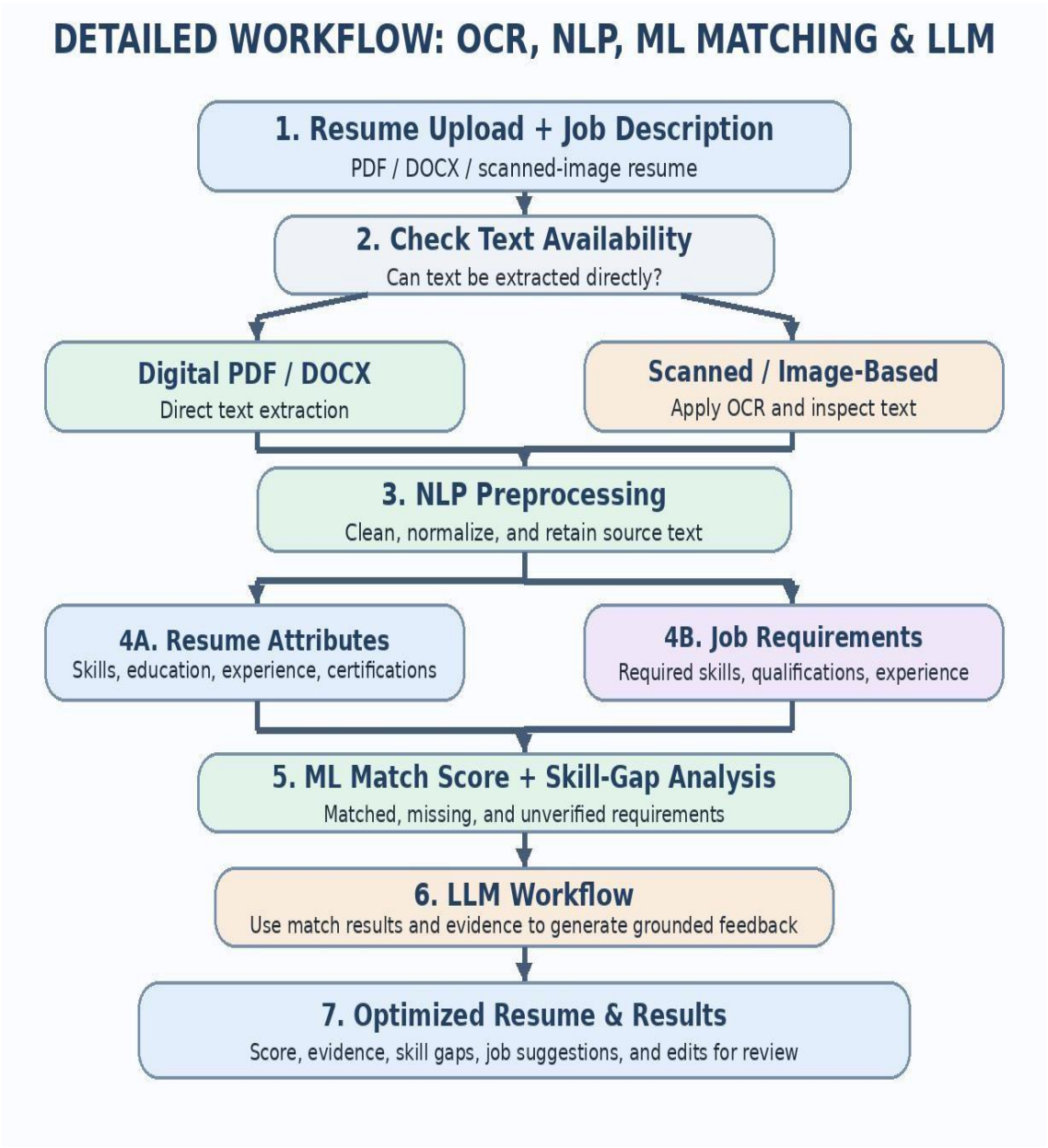


Fig. 3.1: Working of ML-Based ATS Resume Analyzer and Optimizer

3.3 System Architecture

The proposed system consists of several interconnected modules. Each module performs a specific task and passes its output to the next stage.
The main components of the system architecture are as follows:

1. **User Interface Module**
   This module allows users to upload their resumes, enter job descriptions, and view the analysis results.
2. **Text Extraction Module**
   This module extracts text from digital resumes. OCR is used when text must be recognised from scanned or image-based documents.
3. **NLP Processing Module**
   This module preprocesses the extracted text and identifies relevant information such as skills, qualifications, and work experience.
4. **Machine Learning Module**
   This module compares resume information with job requirements using the selected matching method and produces a relevance score or ranking.
5. **Skill Gap Analysis Module**
   This module highlights important job requirements that may be missing or insufficiently represented in the resume.
6. **LLM Feedback Module**
   This module generates suggestions for improving the resume using the resume content, job description, and analysis results as context.
7. **Result Display Module**
   This module presents the analysis results, matching information, identified skill gaps, and improvement suggestions.

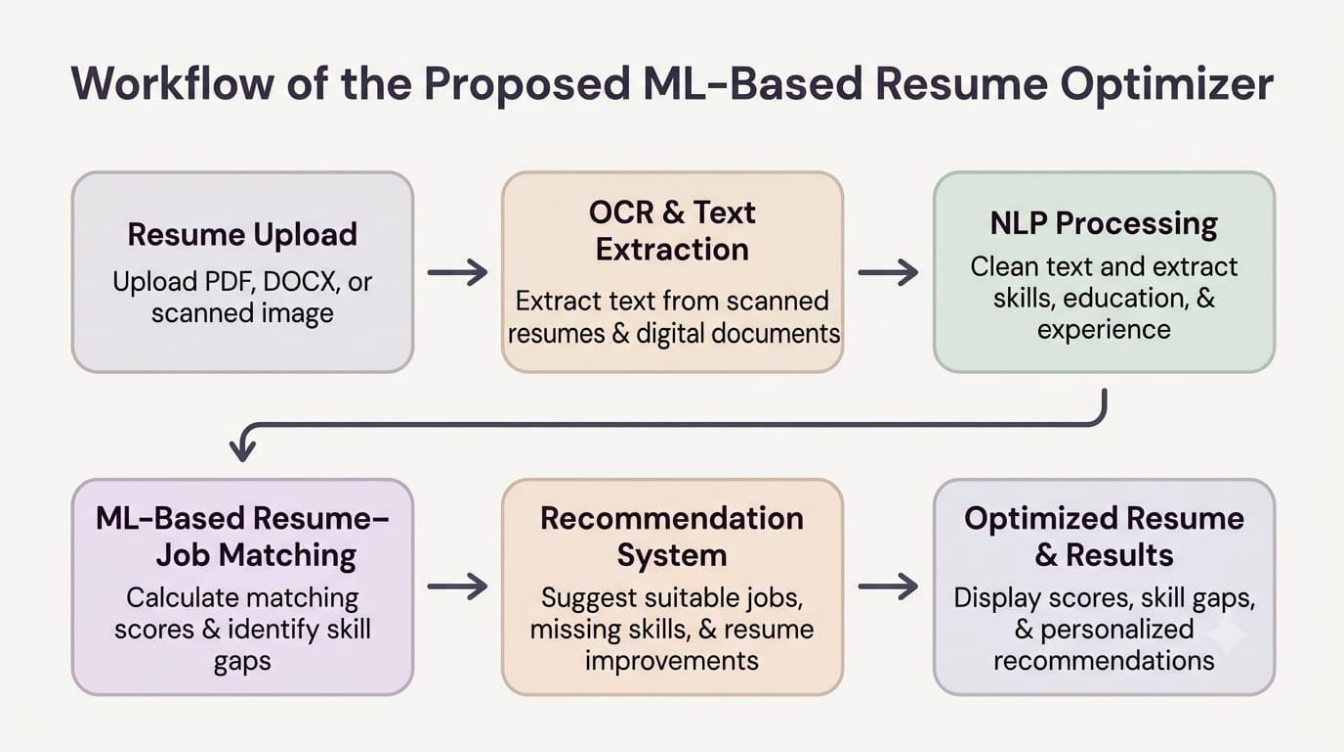


Fig. 3.2: Architecture of ML-Based ATS Resume Analyzer and Optimizer

3.4 Techniques and Technologies Used

The proposed system uses four main technology categories: OCR, NLP, Machine Learning, and Large Language Models.

3.4.1 Optical Character Recognition (OCR)

Optical Character Recognition is a technology used to convert text present in images into machine-readable text.

In the proposed system, OCR is applied when a resume is scanned or contains image-based text. For digital PDFs and other documents with extractable text, direct text extraction can be used instead.

The extracted text is passed to the preprocessing stage for further analysis.

Main applications of OCR in the proposed system:

- Extracting text from scanned resumes.
- Processing image-based resume documents.
- Making recognised text available for NLP processing.

OCR performance may be affected by poor image quality, unusual fonts, complex layouts, and recognition errors.

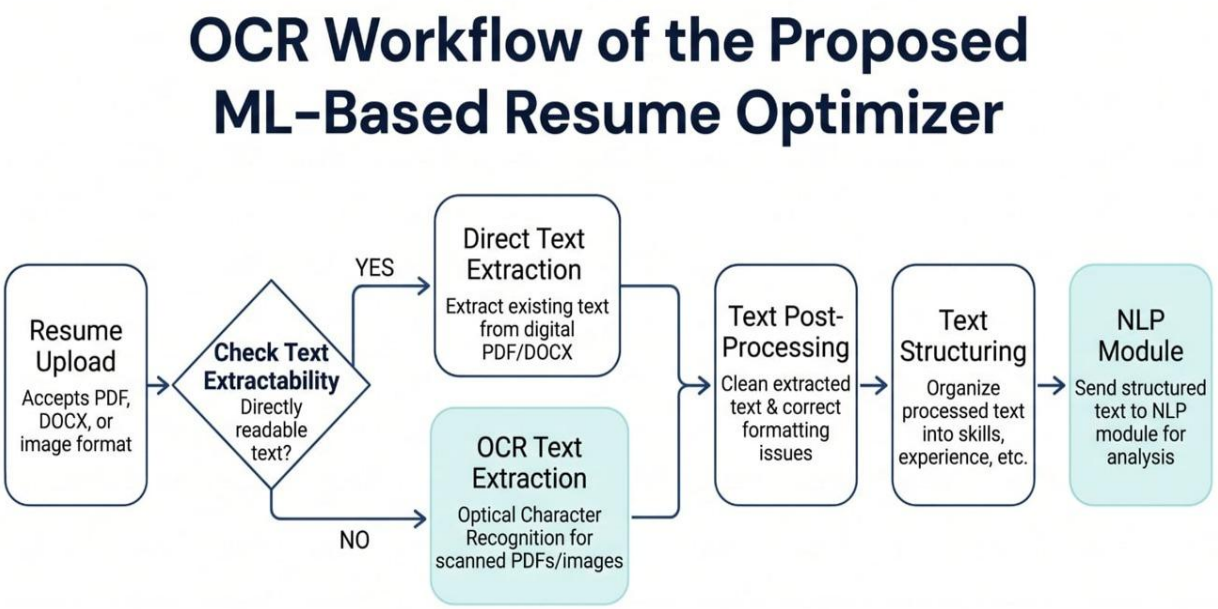


Fig. 3.3: OCR Workflow of the Proposed ML-Based Resume Optimizer

3.4.2 Natural Language Processing (NLP)

Natural Language Processing is a branch of Artificial Intelligence that enables computers to process and analyse human language.

Resumes contain unstructured information written in different formats and styles. NLP techniques help process this information and identify important details.

In the proposed system, NLP is used to:

- Clean and preprocess extracted text.
- Identify technical and professional skills.
- Extract educational qualifications and experience.
- Analyse important terms in job descriptions.
- Organise resume information for further comparison.

NLP helps transform unstructured resume content into information that can be used by the Machine Learning module.

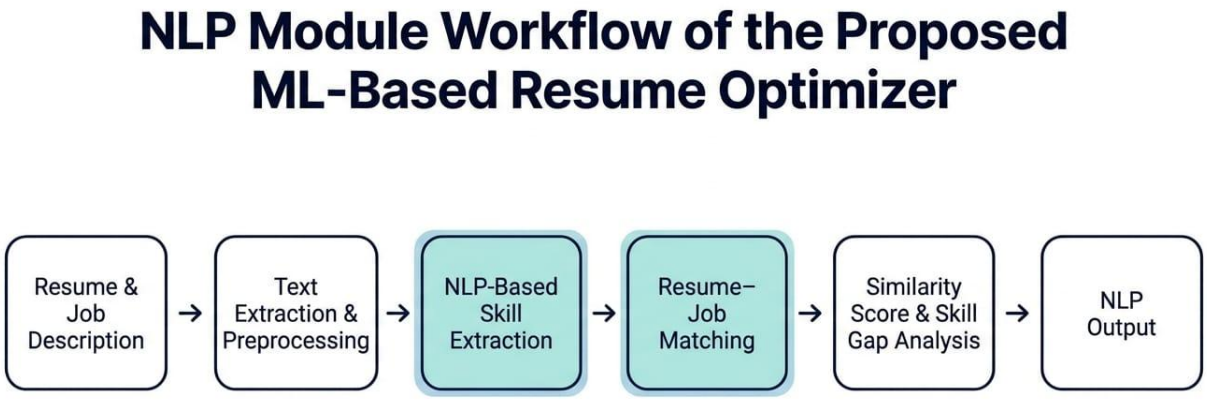


Fig. 3.4: NLP Module Workflow

3.4.3 Machine Learning (ML)

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and use them to make predictions or perform analytical tasks.

In the proposed system, Machine Learning techniques are used to compare extracted resume information with job requirements.

Depending on the selected algorithm, the system may estimate resume–job compatibility or rank resumes according to their relevance to a particular job description.

The main applications of Machine Learning in the proposed system are:

- Comparing resume information with job requirements.
- Estimating resume–job compatibility.
- Supporting resume ranking, if implemented.

Identifying potential mismatches between candidate information and job requirements. The reliability of the results depends on the selected method, input quality, and evaluation process.

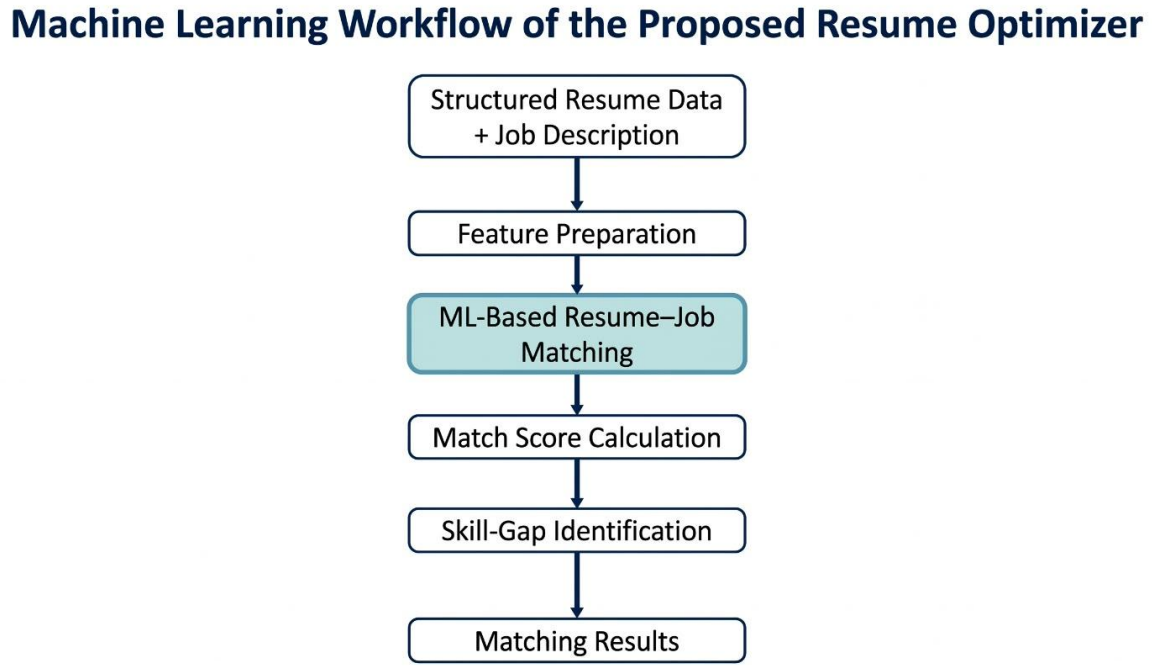


Fig. 3.5: Original Machine Learning Workflow

3.4.4 Large Language Models (LLMs)

Large Language Models are AI models trained to understand and generate human language. In the proposed system, an LLM is used to generate resume improvement suggestions based on the available resume content, job description, and analysis results.

The LLM may suggest clearer wording, better organisation of information, and ways to highlight relevant existing skills and experience.

The main applications of LLMs in the proposed system are:

- Generating personalised resume feedback.
- Suggesting improvements in sentence clarity.
- Helping users present relevant qualifications more effectively.
- Explaining potential skill gaps.
- Providing job-specific resume improvement suggestions.

The generated suggestions must be checked for relevance and factual accuracy. The system should not invent qualifications, skills, certifications, or work experience that the candidate does not possess.



Fig. 3.6: LLM Workflow for Resume Feedback

3.5 Detailed Workflow of the Proposed System

The proposed system integrates OCR, NLP, Machine Learning, and LLM technologies into a single workflow.

First, the user provides a resume and the relevant job description. The system extracts the text directly from digital documents and applies OCR when the resume contains scanned or image-based text.

Next, the extracted text is preprocessed and passed to the NLP module. The NLP module identifies relevant resume information and analyses the job description to determine the required skills and qualifications.

The Machine Learning module then compares the extracted resume information with the identified job requirements. Based on the selected matching method, the system generates a compatibility score or ranking.

The skill gap analysis stage identifies requirements that may be missing or insufficiently represented in the resume. Finally, the LLM uses the available information to generate suggestions for improving the resume.

The system displays the analysis results and suggestions to the user for review.

3.6 Performance Evaluation Metrics

The proposed system should be evaluated using suitable metrics to determine the quality of its output. The metrics selected will depend on the functionality implemented and the availability of verified test data.

3.6.1 OCR Evaluation

OCR performance can be evaluated by comparing the extracted text with manually verified text from the original document.

This comparison helps identify recognition errors and missing text. The evaluation should use representative scanned resumes with different layouts and image qualities.

3.6.2 NLP Evaluation

NLP-based information extraction can be evaluated by comparing the skills and other details identified by the system with manually verified information.

Precision, recall, and F1-score may be used when suitable labelled test data is available.

- **Precision:** Measures how many identified items are correct.
- **Recall:** Measures how many relevant items the system successfully identifies.
- **F1-score:** Combines precision and recall into a single measure.

3.6.3 Resume–Job Matching Evaluation

The matching module can be evaluated using representative resume and job description pairs with manually assessed relevance.

Suitable evaluation methods may include Precision@K, Recall@K, or ranking-based metrics when the system ranks multiple resumes. The evaluation method should match the actual functionality implemented.

3.6.4 LLM Feedback Evaluation

LLM-generated feedback can be reviewed for relevance, clarity, factual consistency, and usefulness.

evaluation should check whether the suggestions are supported by the resume and job description and whether they avoid introducing false information.

3.7 Challenges and Limitations

The proposed system may face the following challenges and limitations.

1. **OCR Errors:**
2. Poor-quality scans, unusual fonts, and complex resume layouts mayresult in incorrect text extraction.
3. **Variations in Resume Format:**
4. Resumes differ in structure, formatting, and terminology, which can make information extraction more difficult.
5. **NLP Limitations:**
6. NLP techniques may fail to recognise some technical terms, abbreviations, or context-dependent expressions.
7. **Matching Score Limitations:**
8. A high matching score does not guarantee that a candidate is suitable for a job or will be selected by a recruiter.
9. **Data Quality:**
10. Incomplete or inaccurate resume information and job descriptions may affect the analysis.
11. **LLM Reliability:**
12. An LLM may generate irrelevant or unsupported suggestions, so its output must be checked.
13. **Bias:** Biased training data or evaluation criteria may lead to unfair results.
14. **Privacy and Security:**
15. Resumes contain personal information that must be handled securely and used appropriately.
16. **Computational Requirements:**
17. Some NLP models and LLM services may require additional computing resources or internet connectivity.
18. **Need for Human Review:**

19. The proposed system should support resume improvement and analysis rather than replace human judgement in recruitment decisions.

Overall, the proposed system demonstrates a possible integration of OCR, NLP, Machine Learning, and LLM technologies for resume analysis and optimization. Its effectiveness must be established through implementation and systematic evaluation.

Chapter 4: Advantages And Disadvantages

4.1 Advantages

1. Automated Resume Analysis: The proposed system automates the analysis of resumes against job descriptions, reducing the effort required for manual review.
2. Efficient Text Extraction: OCR can extract text from scanned or image-based resumes, while direct text extraction can be used for suitable digital documents.
3. Improved Information Extraction: NLP techniques help identify relevant skills, educational qualifications, work experience, projects, and certifications from resume content.
4. Resume–Job Description Matching: Machine Learning techniques support the comparison of resume information with job requirements and help estimate their relevance.
5. Skill Gap Identification: The system can highlight job requirements that are missing or insufficiently represented in a candidate's resume.
6. Personalized Resume Suggestions: LLMs can generate suggestions to improve resume clarity, wording, and presentation according to a specific job description.
7. Time Efficiency: Automated processing can reduce the time spent manually identifying and comparing relevant information across resumes.
8. Consistent Analysis: Applying a defined workflow to different resumes can support a more consistent approach to information extraction and matching.
9. Better Resume Presentation: The system can suggest ways to organise existing skills and experience more clearly without changing the candidate's actual qualifications.
10. Support for Job Seekers: The system can help candidates understand job requirements, review their resume content, and identify areas for improvement.

4.2 Disadvantages

1. OCR Recognition Errors: Poor image quality, unusual fonts, and complex layouts can cause OCR to extract incorrect or incomplete text.
2. NLP Limitations: NLP techniques may fail to identify certain technical terms, abbreviations, or skills expressed using unfamiliar terminology.
3. Matching Score Limitations: A high resume–job compatibility score does not guarantee that a candidate is suitable for a position or will be selected for an interview.
4. Dependence on Input Quality: Incomplete resumes or unclear job descriptions can reduce the reliability of the analysis.
5. LLM-Generated Inaccuracies: LLMs may produce irrelevant or unsupported suggestions. Generated content must be checked against the original resume and job description.
6. Potential Bias: Biased datasets or poorly designed evaluation criteria may lead to unfair or inconsistent results.
7. Privacy and Security Risks: Resumes contain personal information that must be protected against unauthorised access and inappropriate use.
8. Implementation Complexity: Integrating OCR, NLP, Machine Learning, and LLMs requires appropriate technical knowledge, testing, and maintenance.

9. Computational and Financial Costs: Some models and cloud-based LLM services may require additional computing resources, internet access, or usage fees.
10. Need for Human Review: The system cannot fully assess every aspect of a candidate's abilities, experience, or potential. Its results should support human judgement rather than replace it.

Chapter 5: Conclusion

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system designed to simplify resume analysis and improve compatibility with Applicant Tracking Systems (ATS). It aims to help job seekers understand how effectively their resumes match the requirements of a particular job description.

The proposed system integrates Optical Character Recognition (OCR), Natural Language Processing (NLP), Machine Learning (ML), and Large Language Models (LLMs). OCR can extract text from scanned resumes, NLP can identify important information such as skills, qualifications, and experience, and ML techniques can support resume–job description matching and scoring. An LLM can then generate personalized suggestions to improve resume content and presentation.

The system is intended to identify relevant skills, highlight missing job-related keywords, and provide recommendations for improving resume structure and relevance. By automating these tasks, it may reduce the effort required for initial resume analysis and help candidates prepare resumes tailored to specific job opportunities.

However, the effectiveness of the system depends on the quality of the input documents, the accuracy of text extraction, the reliability of the matching techniques, and the quality of generated suggestions. Privacy, potential bias, and incorrect recommendations must also be considered. Therefore, automated scores and suggestions should support, rather than replace, human judgment.

In conclusion, the proposed ML-Based ATS Resume Analyzer and Optimizer offers a practical approach to combining modern AI technologies for resume evaluation and improvement. Its actual performance and usefulness must be established through implementation, testing, and evaluation using suitable resumes and job descriptions.

Chapter 6: Project Initiation

6.1 Problem Statement

Recruitment involves analyzing resumes and comparing candidates' qualifications, skills, and experience with the requirements of a particular job. Manually reviewing multiple resumes can be time-consuming, and resumes may contain different formats, layouts, and descriptions of similar skills. Conventional keyword-based screening may also overlook relevant information when the wording used in a resume differs from that in the job description.

The proposed **ML-Based ATS Resume Analyzer and Optimizer** aims to address these challenges by automating resume information extraction, job-description analysis, resume matching, and the identification of skill gaps. Natural Language Processing (NLP) is used to process text and identify relevant information, while Machine Learning (ML) techniques support the comparison and ranking of resume content against job requirements. Optical Character Recognition (OCR) can be used when text needs to be extracted from scanned or image-based resumes. A Large Language Model (LLM) may generate explanations and resume-improvement suggestions based on the extracted information and matching results.

The main problem is to design a workflow that provides useful, understandable, and relevant resume-matching results while maintaining accuracy, privacy, and fairness. The system is intended to support job seekers and assist resume evaluation; its scores should not be treated as guarantees of selection or interview calls.

6.2 Scope and Objectives

6.2.1 Scope

The scope of the proposed project includes the following functions:

1. Accepting resumes in supported digital document formats.
2. Extracting text directly from digital documents and applying OCR when required for scanned resumes.
3. Processing resume text using NLP techniques.
4. Identifying skills, educational qualifications, work experience, and other relevant information.
5. Extracting required skills and qualifications from a job description.
6. Comparing resume information with job requirements using suitable matching techniques.
7. Calculating a resume–job matching score using a defined method.
8. Identifying missing or underrepresented skills and keywords.
9. Generating suggestions for improving resume relevance, clarity, and presentation.
10. Evaluating the proposed workflow using suitable test data and task-appropriate metrics.
11. The project focuses on resume analysis and optimization for a given job description. The system will not make final recruitment decisions, guarantee ATS acceptance, or recommend adding qualifications and skills that the candidate does not actually possess.

6.2.2 Objectives

The main objectives of the project are:

1. To study existing techniques for resume parsing, skill extraction, and resume–job matching.
2. To design a workflow for extracting information from digital and scanned resumes.
3. To apply NLP techniques to identify important resume attributes and job requirements.
4. To investigate ML-based methods for resume matching and ranking.
5. To calculate a meaningful matching score and identify possible skill gaps.
6. To generate understandable, job-specific resume-improvement suggestions.
7. To evaluate the accuracy and usefulness of the proposed methods using suitable metrics.
8. To consider privacy, fairness, explainability, and human review during system design.

6.3 Stakeholders

The stakeholders involved in the proposed project are listed below.

Job Seekers: Provide resumes and job descriptions, review matching results, and decide which suggestions to apply.

Recruiters: May use the analysis as supporting information during initial resume screening.

Project Developers: Design the system workflow, integrate the required technologies, and conduct testing.

Project Guide: Provides guidance on project planning, technical design, documentation, and evaluation.

Educational Institution: Defines academic requirements and evaluates the seminar report and presentation.

System Administrator: Where applicable, manages system access, configuration, and data security.

6.4 Seminar Discussion Plan

The following five-week plan is proposed for literature review, system design, evaluation planning, and seminar report preparation. The schedule can be adjusted according to the institute's academic calendar and project requirements.

Week	Planned Activity	Expected Output
Week 1	Define the problem statement, identify project objectives, and study the supplied research papers.	Problem statement, initial notes, and source list.
Week 2	Compare existing approaches for resume parsing, skill extraction, matching, ranking, and resume optimization.	Literature comparison and identification of limitations.
Week 3	Design the proposed system architecture, workflow, input requirements, matching approach, and output format.	System architecture diagram and proposed methodology.
Week 4	Define test data, evaluation criteria, suitable metrics, privacy measures, and limitations.	Evaluation plan and testing checklist.
Week 5	Prepare the seminar report, organize figures and tables, verify references, and prepare presentation slides.	Final seminar report and seminar presentation.

Table 6.1: Five-Week Seminar and Project-Initiation Plan

The planned activities describe the intended project schedule. Implementation and experimental results should be reported only after the corresponding work has been completed.

6.5 Platform / Equipment / Resources Identification

The proposed project requires hardware, software, document-processing tools, suitable test data, and evaluation resources.

Resource	Purpose / Requirements
Laptop / Computer	Used for development, document processing, testing, and report preparation.
Operating System	Windows, Linux, or another supported development environment.
Programming Language	Python for text preprocessing, NLP, matching, integration, and evaluation.
Data Processing Libraries	pandas and NumPy for organizing and processing structured data.
Machine Learning Library	scikit-learn for baseline matching techniques, similarity calculations, and evaluation metrics.
NLP / Embedding Tools	Suitable NLP libraries or sentence-embedding models for skill extraction and semantic comparison, depending on the selected approach.
Document Extraction Tools	PDF and DOCX parsing libraries for extracting text from supported resume formats.
OCR Tool	Used when text must be extracted from scanned or image-based resumes.

LLM Service or Model	Optional component for generating evidence-based explanations and resume-improvement suggestions.
Development Environment	Visual Studio Code, Jupyter Notebook, or another suitable coding environment.
Version Control	Git or a similar tool for tracking source-code changes, if used.
Dataset	Permitted and appropriately anonymized resumes and job descriptions for testing.
Evaluation Resources	Annotation sheets, test cases, evaluation scripts, and an error-review checklist.
Privacy and Security Measures	Access control, data minimization, anonymization, and appropriate retention practices.
Documentation Tools	Word processor and presentation software for preparing the report and seminar slides.

Table 6.2: Platform, Equipment and Resources

The final selection of tools and libraries will depend on the implementation approach and available resources. The proposed system should be tested with suitable data before its accuracy or effectiveness is claimed.

Chapter 7: Seminar Presentation

S.S.V.P.S's B. S. Deore Polytechnic, Dhule Department of Computer Engineering *Academic Year : 2026-2027*		
SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5 K	**SPI – 315003**	**Date : 17/10/2026**
Topic : ML – Based ATS Resume Optimizer		

Roll No.	Name of Student
66	Patil Manasi Vijay
92	Singh Digvijay Manoj
110	Wagh Tanishka Mahesh
111	Wani Roshani Rupak

Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of Problem Statement.
   - 6. 2 Scope and Objectives.
   - 6. 3 Stake holder
   - 6. 4 Seminar Discussion Plan
   - 6.5 Platform/ Equipment/ Resources identification.

- Bibliography
- References

Abstract

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system that analyzes a resume against a selected job description. Text is extracted directly from digital resumes; Optical Character Recognition (OCR) is applied only when a resume is scanned or image-based. Natural Language Processing (NLP) identifies skills, education, experience, projects, and certifications in the resume, and required skills and criteria in the job description. A Machine Learning (ML) matching module estimates compatibility and finds matched, missing, and unverified requirements. A Large Language Model (LLM) then uses these results and the source evidence to explain skill gaps and suggest resume improvements, which the user reviews before use. The system is at the proposal stage; implementation and testing are future work.

1. Introduction

In today's competitive job market, candidates apply for many jobs, but they often find it difficult to prepare a resume that matches specific job requirements. Employers also use Applicant Tracking Systems (ATS) to screen and filter a large number of resumes. We chose the ML-Based Resume Optimizer topic because many candidates struggle to identify missing skills, improve resume content, and understand how well their resumes match a particular job description.

The proposed ML-Based Resume Optimizer aims to help candidates improve their resumes through the following stages:

- **OCR** : Extracts text from scanned or image-based resumes.
- **NLP** : Identifies important information such as skills, education, and work experience.
- **ML** : Calculates resume-job matching scores and identifies skill gaps.
- **Resume Optimization** : Provides suggestions to improve resume content and highlight relevant skills.

This project can help candidates prepare better-targeted resumes, identify areas for improvement, and make the job application process more effective. The matching score provides guidance only and does not guarantee ATS selection or employment.

2. Literature Survey

Sr. No.	Category	Paper Title
1	**OCR (STAGE 1)**	Resume Information Extraction via Post-OCR Text Processing
2	**NLP (STAGE 2)**	Technical Skills Information Extraction from Resumes Using Advanced Natural Language Processing Model with Transfer Learning
3	**NLP (STAGE 2)**	Resume Screening with Natural Language Processing (NLP)
4	**NLP (STAGE 2)**	Smart Hiring: An NLP and Ensemble-Based Approach to AI-Driven Resume Screening
5	**ML (STAGE 3)**	Resume Screening Using Machine Learning
6	**ML (STAGE 3)**	Automated Resume Screening Using Machine Learning: An Ensemble-Based Approach for Efficient Candidate Selection
7	**ML (STAGE 3)**	Design and Development of Machine Learning Based Resume Ranking System
8	**LLM (STAGE 4)**	Optimizing Resume Design for ATS Compatibility: A Large Language Model Approach
9	**LLM (STAGE 4)**	Resume Analyzer Using LLM
10	**LLM (STAGE 4)**	An LLM-Based Resume Intelligence System for Recruiters and Job Seekers

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3. Information to Seminar topic

The proposed system follows four stages: OCR → NLP → Machine Learning → LLM. Each stage performs a specific task to analyze resumes and compare them with job requirements.

1. Optical Character Recognition (OCR) : OCR extracts text from scanned or image-based resumes. For digital PDF and DOCX files containing readable text, direct text extraction is used instead of OCR. This avoids unnecessary processing and supports different resume formats.

2. Natural Language Processing (NLP) : NLP cleans and normalizes the extracted text. It identifies relevant information from two sources: the resume, including skills, education, experience, projects, and certifications; and the job description, including required skills, qualifications, and experience. This converts unstructured text into information suitable for comparison.

3. Machine Learning (ML) : The ML stage compares the structured resume information with the job requirements. The proposed outputs include an estimated compatibility score, matched requirements, missing requirements, unverified requirements, and skill-gap analysis. These results help candidates understand how closely their resumes align with a particular job.

4. Large Language Model (LLM) : The LLM uses the matching results and supporting resume evidence to explain skill gaps and suggest clearer wording for existing qualifications and experience. It must not invent skills or qualifications. Suggestions should be checked against the original resume and reviewed by the user before use.

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3.1 Proposed System Architecture

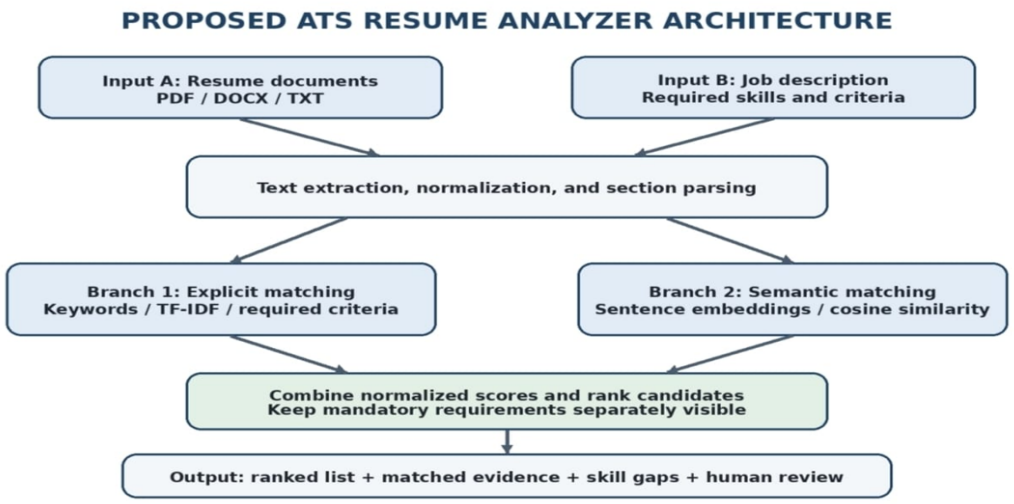


Fig. 3.1: Proposed ATS Resume Analyzer Architecture

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3.2 Overall System Workflow

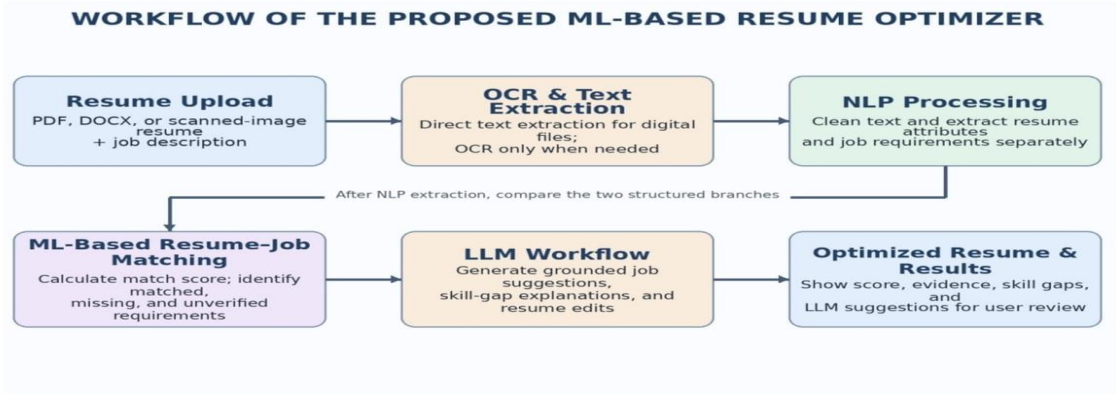


Fig. 3.2: Workflow of the Proposed ML – Based Resume Optimizer

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3.3 Detailed Working of the System

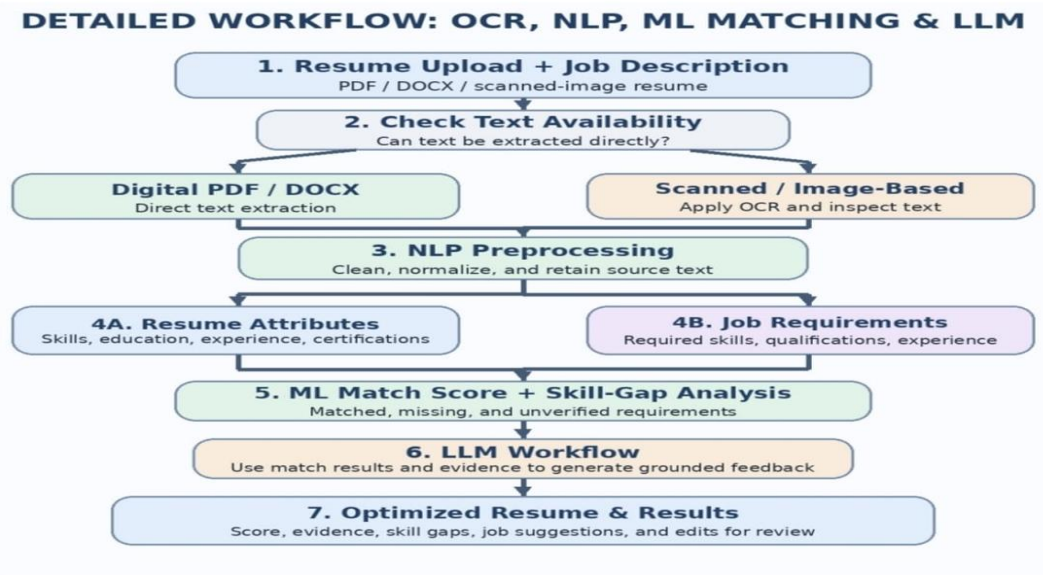


Fig. 3.3: Detailed Working of the System

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3.4 OCR Based Resume Workflow

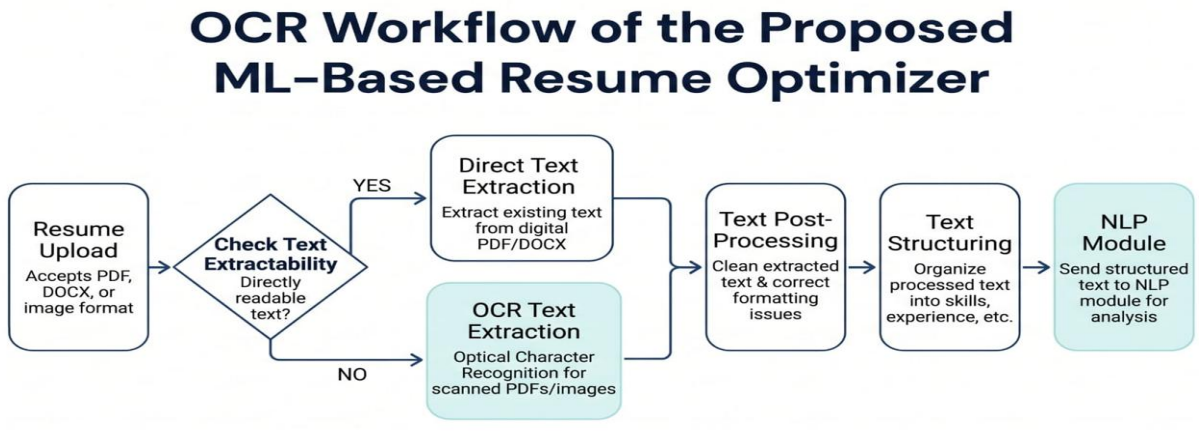


Fig. 3.4: OCR Workflow of the Proposed ML – Based Resume Optimizer

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3.5 NLP Module Workflow

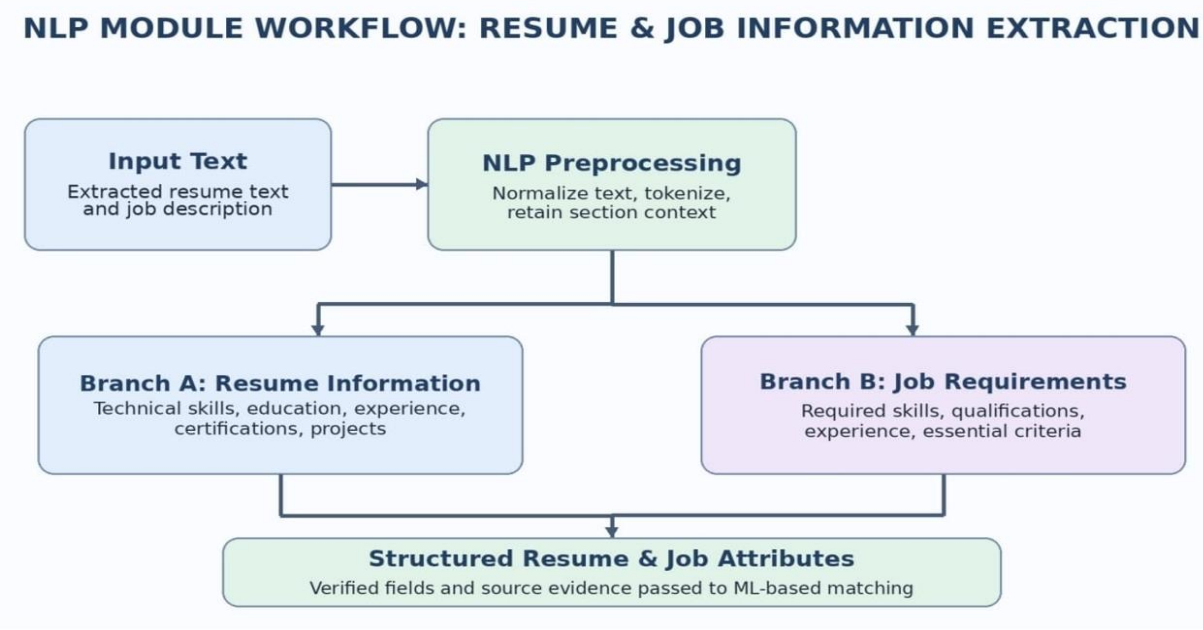


Fig. 3.5: NLP Workflow : Resume & Job Information Extraction

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3.6 ML System Workflow

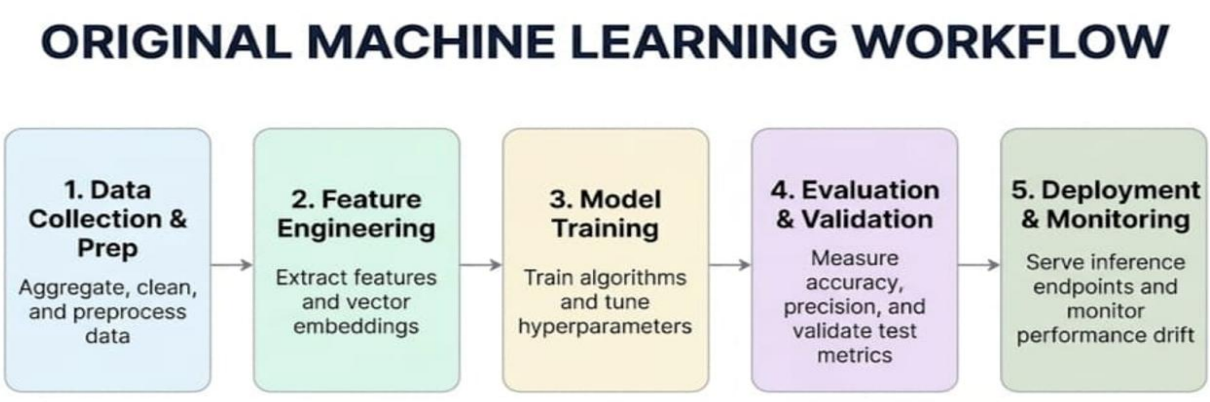


Fig. 3.6: Original Machine Learning Workflow

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4. Advantages / Disadvantages

Advantages

- Structured extraction of resume information.
- Support for scanned resumes through OCR.
- Comparison of the resume with job requirements.
- Identification of potential skill gaps.
- Personalized, evidence-based feedback.
- Reduced manual effort.

Disadvantages

- OCR errors on poor-quality scans.
- NLP extraction errors; results depend on data quality.
- Limitations of the matching model.
- LLM may generate unsupported claims.
- Privacy and security of personal data.
- Human review is always needed.

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5. Conclusion

The ML-Based ATS Resume Analyzer and Optimizer is a proposed system designed to help candidates improve their resumes and understand how well they match specific job requirements. The proposed workflow consists of four stages: OCR extracts text from scanned or image-based resumes when required, NLP processes and organizes resume and job information, Machine Learning estimates compatibility and identifies potential skill gaps, and an LLM explains the results and suggests resume improvements. These features are proposed and have not yet been experimentally evaluated. Therefore, no claims are made regarding system accuracy or superior performance compared with other models. The compatibility score is intended to provide guidance only and does not guarantee ATS acceptance or employment.

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6. Project Initiation

Project Title: ML-Based Resume Optimizer

Project initiation defines the main problem, objectives, scope, stakeholders, and resources required for the proposed system.

The proposed project aims to develop a system that analyzes resumes, compares candidate information with job requirements, identifies skill gaps, and recommends suitable job roles and resume improvements.

The project planning phase helps identify the necessary technologies, tools, and resources required for implementation.

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6. 1 Description of Problem Statement

Problem

- Job seekers often cannot tell how well a resume matches a job description or which requirements are missing or weakly documented.
- Resumes come in many formats, including scanned images, which makes information extraction difficult.
- Manual evaluation is slow, and simple keyword matching can miss related wording.

Problem Statement

To design a system that extracts information from resumes and job descriptions, estimates compatibility, identifies potential skill gaps, and gives evidence-based resume-improvement suggestions for user review.

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6. 2 Scope and Objectives

Scope of the Project

- Accept resumes in supported document and image formats.
- Extract text using OCR when required.
- Identify skills, education, and work experience using NLP.
- Compare resume information with job descriptions.
- Identify skill gaps and suggest suitable job roles.
- Provide recommendations for resume improvement.

Objectives

- Automate resume information extraction.
- Analyze compatibility between resumes and job requirements.
- Help candidates identify missing skills.
- Support better resume preparation and job-role selection.

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6. 3 Stake holder

The main stakeholders of the proposed ML-Based Resume Optimizer are :

1. **Job Seekers:** Upload resumes, review matching results, and improve their resumes.
2. **Recruiters and HR Teams:** Can use matching results to support initial resume screening.
3. **Project Developers:** Design, develop, test, and maintain the system.
4. **Educational Institutions:** Can use skill-gap information to guide students in developing relevant skills.
5. **System Administrator:** Manages system access, operation, and data security.

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6. 4 Seminar Discussion Plan

The seminar discussion will cover the following topics :

1. Introduction to the ML-Based Resume Optimizer.
2. Literature survey and existing resume-screening approaches.
3. Overview of OCR, NLP, Machine Learning, and Recommendation Systems.
4. Proposed system architecture and working process.
5. Advantages, disadvantages, and applications.
6. Project scope, objectives, and required resources.
7. Conclusion and future improvements.

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6.5 Platform/ Equipment/ Resources

Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM; 8 GB or more recommended.
- Internet connection.

Software Requirements

- Operating System: Windows or Linux.
- Programming Language: Python.
- Development Environment: VS Code or Jupyter Notebook.
- NLP and ML Libraries: NLTK, spaCy, or scikit-learn.
- OCR Tool: Tesseract OCR or a suitable OCR library.

Other Resources

- Sample resumes for testing.
- Job descriptions for different roles.
- Resume–job matching data for evaluation.

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Thank you

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Formative Assessment

Roll No	Enrollment No	Name	Exam Seat No.
66	24610960246	Patil Manasi Vijay	388553
92	24610960279	Singh Digvijay Manoj	399579
110	24610960298	Wagh Tanishka Mahesh	388597
111	24610960299	Wani Roshani Rupak	388598

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388553	399579	388597	388598
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign:

Name:Mr.Nishad.D.Patel

(Course Experts/s)